Mid Term Examination, Spring 2024
Department of Computer Science and Engineering (CSE)
Level-2, Term-II
Course Code: MATH 221
Course Title: MATH IV (Probability & Statistics)

Notes: **Time : 1 hr.**
Marks : 60

a. Figure on the right of each question indicate marks for respective question
b. **Answer any 02 (Two) of the following 03 (Three) questions including Q -03.**

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1. a. Compute the Geometric mean and Arithmetic mean (Short cut method) for the following data set: 13

Class Interval	20-25	25-30	30-35	35-40	40-45	45-50	50-55
Frequency	10	12	8	20	11	4	5

b. Describe the relation between arithmetic mean, geometric mean and harmonic mean. 07

c. Construct a Cumulative Frequency polygon and histogram corresponding to the frequency distribution of the following data. 10

Class boundary	48.5-58.5	58.5-68.5	68.5-73.5	73.5-78.5	78.5-98.5
Frequency	4	8	5	5	28

2. a. A population consists of the five numbers 3, 5, 7, 8, 10. Consider all possible samples of size two which can be drawn without replacement from this population. Compute 15

(1) The mean of the population
(2) The standard deviation of the population
(3) The mean of sampling distribution of means
(4) The standard deviation of the sampling distribution of means.

b. The table below gives data on the heights, in cm, of 51 children. 15

Class interval	$140 \le h < 150$	$150 \le h < 160$	$160 \le h < 170$	$170 \le h < 180$
Frequency	6	16	21	8

Obtain the median and the mode, for the distribution of height.

3. a. A disease is known to be present 2% of a population. A test is developed to help determine whether or not someone has the disease. Given that a person has the disease, the test is positive with probability 0.95. And given that a person does not have the disease, the test is positive with probability 0.03. 12

(i) Draw a tree diagram to represent this information.

(ii) A person is selected at random from the population and tested for this diseases. Find the probability that the test is positive.

(iii) A doctor randomly selects a person from the population and tests him for the disease. Given that the test is positive. Find the probability that he does not have the disease.

(iv) Discuss the usefulness of this test.

b. A pair of dice is thrown. Compute the probability that the sum of the points on the two dice is 10 or greater if a 5 appears on the first. 07

c. Three coins are tossed. Show that the events " Heads on the first coin" and "Tails on the last two" are independent. 05

d. A leap year consists of 366 days with 29 days in February. If a leap year is selected at random, what is the probability that the selected leap year will consist of 53 Saturdays? 06

বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি, কুমিল্লা
BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Mid Term Examination, Spring 2024

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE-225

Course Title: Numerical methods

Credit Hour: 03

Exam Duration: 1 Hour

Full Marks: 60

Notes:

a. Figure on the right of each question indicate marks for respective question.
b. Answer ***any TWO*** questions out of **THREE**.

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1. a. How do round-off errors and truncation errors impact the accuracy of numerical methods? 10
   b. Convert $(43.375)_{10}$ to binary. 10
   c. Find out the precision and accuracy for the following numbers 10

      i) 80.500 ii) 0.0080775 iii) 9000 iv) 05.00023410 v) 55

2. a. Subtract 0.994576E-5 from 0.999658E-3 following subtraction algorithm of computer arithmetic. 10
   b. Write multiplication algorithm for computer arithmetic. 10
   c. Find the round off and chopping error in storing the number 752.6835 using a four-digit mantissa. 10

3 a. Using the bisection method, determine the root of the following function $f(x)=3x-\cos x-1$ within the interval [0,1] to a tolerance of 10^{-3}. 15
   b. By utilizing the false position method, compute the root of the following function 15
   $2e^{x}\sin x=3$ within the interval [0,1] and correct it to three decimal places with iterations.

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Mid Term Examination, Spring 2024

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE 221

Course Title: Computer Algorithms and Complexity

Credit Hour: 3.0

Exam Duration: 1 hour

Full Marks: 60

Notes:

a. Figure on the right of each question indicates marks for respective question.

b. Answer ***any TWO*** questions out of **THREE**.

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1. **a.** What is an algorithm? Briefly explain the essential characteristics of algorithm. **10**

   b. Consider the following list of values: **3, 7, 4, 9, 5, 2, 6, 1**. Perform the insertion sort algorithm step by step to sort these values in ***ascending order***. Show each step of the algorithm, including the intermediate state of the list after each insertion. **10**

   c. Write down the first Partition of the given data using the Quick sort algorithm: **10**

   40, 30, 50, 22, 60, 55, 77, 55

2. **a.** Construct a Step Table for the following algorithm. **10**

```
Algorithm RSum(a, n)
{
    if (n<=0) then
        Return 0;
    else return
        RSum (a, n-1) + a[n];
}
```

   b. Verify that the constructed **5x5** magic square satisfies the magic square properties by showing that the sums of all rows, columns, and both diagonals are equal to the magic constant. **10**

   c. Write the pseudo code for Bubble sort. **15**

3. **a.** Write the Pseudocode for Stack Insertion, Deletion, and Display operation. **15**

   b. Sort the following data using the heap sort algorithm:

   40, 30, 50, 22, 60, 55, 77, 35

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Mid Term Examination, Spring-2024

Department of Computer Science and Engineering

Level- 2 Term- II

Course Code: CSE 223

Course Title: Digital Logic Design

Credit Hour: 3

Exam Duration: 1 hour

Full Marks: 60

Notes:

a. Figure on the right of each question indicate marks for respective question.

b. Answer ***any TWO*** questions out of **THREE**.

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1. a. A company has four shareholders, A, B, C & D. A owns 30 shares, B 40 shares, C 10 shares & D 20 shares. In order for a particular measure in the shareholders meeting to be passed, shareholders holding more than two-thirds of the total shares should vote for it. Assuming that on the board room each shareholder has a switch which he switches on if he votes 'yes', Design a switching circuit with diagram which will ring a bell if measure is passes. 15

   b. Ross & Rachel have two children, Ben and Emma. When eating out they will go to a restaurant that serves only fish or one that serves only chicken. Before going out, the family votes to decide on the restaurant. The majority wins, except when Mom and Dad agree, and in that case they win. Any other tie votes produce a trip to the chicken restaurant. Design a logic circuit with diagram that will automatically select the restaurant when everyone votes. [1 represent a vote cast for fish & 0 represent a vote cast for chicken] 15

2. a. Find out the conversion result of 09

   (last 6 digit of your ID) + $(7354)_8$ + $(10110101)_2$=$(?)_{16}$

   b. With appropriate Boolean algebra proof X+XY=X. 09

   c Draw the CMOS NAND gate with proper truth table. 12

3. a. Draw the simplified circuit for the following function using K-map: 12+3

   $f(a,b,c,d) = \sum m\,(1,2,6,7,8,13\text{-}15)+d(0,3,12)$

   b. Draw the simplified circuit for the following function using K-map: 12+3

   $f(a,b,c,d) = \prod M\,(3,5,7,8,10,11,12,13)$

SOME IMPORTANT LAWS TO SIMPLIFY LOGIC FUNCTIONS

Law/Theorem	Law of Addition	Law of Multiplication
Identity Law	$x + 0 = x$	$x \cdot 1 = x$
Complement Law	$x + x' = 1$	$x \cdot x' = 0$
Idempotent Law	$x + x = x$	$x \cdot x = x$
Dominant Law	$x + 1 = 1$	$x \cdot 0 = 0$
Involution Law	$(x')' = x$	
Commutative Law	$x + y = y + x$	$x \cdot y = y \cdot x$
Associative Law	$x+(y+z) = (x+y)+z$	$x \cdot (y \cdot z) = (x \cdot y) \cdot z$
Distributive Law	$x \cdot (y+z) = x \cdot y+x \cdot z$	$x+y \cdot z = (x+y) \cdot (x+z)$
Demorgan's Law	$(x+y)' = x' \cdot y'$	$(x \cdot y)' = x' + y'$
Absorption Law	$x + (x \cdot y) = x$	$x \cdot (x + y) = x$